Special Numbers

Year 4 · **Lesson Worksheet**

Worked Examples

Square numbers & roots: Is 81 a square number? If so, find its square root.

Answer: Yes: 9 × 9 = 81, so 81 is a square number and √81 = 9.

Triangular numbers: Find the first five triangular numbers and describe the pattern used to generate them.

Answer: 1, 3, 6, 10, 15 — each triangular number is the sum of consecutive whole numbers starting at 1 (e.g. the 4th is 1+2+3+4 = 10), so each term adds one more than the last addition.

Cube numbers & sum of odd numbers: Show that the sum of the first 4 odd numbers (1+3+5+7) equals a square number.

Answer: 1 + 3 + 5 + 7 = 16 = 4^2, confirming that the sum of the first n odd numbers always equals n^2.

Square & Cube Basics (15 questions)

1. Write down the first six square numbers.
2. What is 7^2?
3. What is the square root of 49?
4. What is 2^3 (2 cubed)? A) 4 B) 6 C) 8 D) 10 E) 12
5. What is the cube root of 27?
6. Is 36 a square number? Explain how you know.
7. Write down the first four triangular numbers.
8. What is the 5th triangular number? A) 10 B) 12 C) 15 D) 18 E) 20
9. What is 10^2 ?
10. Which of these is a cube number? A) 16 B) 25 C) 36 D) 64 E) 90
11. Select ALL that apply: which of these numbers are square numbers? A) 9 B) 16 C) 20 D) 25 E) 32
12. What is the square root of 100?
13. What is 4^3 (4 cubed)?
14. Select ALL that apply: which of these numbers are triangular numbers? A) 6 B) 8 C) 10 D) 14 E) 15
15. What is $3^2 + 2^3$? A) 15 B) 16 C) 17 D) 18 E) 19

Patterns & Classifying (15 questions)

1. A number is both a square number and an even number, and it lies between 50 and 100. What is the number?
2. The sum of the first 3 odd numbers is 1 + 3 + 5 = 9. Use this pattern to write the sum of the first 6 odd numbers without adding them one by one.
3. What is the 7th triangular number?

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4. Which of these numbers is a triangular number? A) 21 B) 22 C) 23 D) 24 E) 25

5. Select ALL that apply: which of these statements about 64 are true? A) 64 is a square number. B) 64 is a cube number. C) The square root of 64 is 8. D) The cube root of 64 is 4. E) 64 is an odd number.

6. Find two triangular numbers that add together to make 25.

7. What is the difference between the 6th and 5th square numbers?

8. A number is a cube number greater than 30 and less than 100. What is the number? A) 27 B) 36 C) 49 D) 64 E) 81

9. The 8th triangular number is 36. What is the 9th triangular number? (Hint: each new term adds one more than the previous term added.)

10. Select ALL that apply: which of these numbers are both square numbers and triangular numbers? A) 1 B) 9 C) 16 D) 36 E) 49

11. What is the sum of the first 5 odd numbers? What type of special number does this give?

12. Which square number, when you find its square root, gives an odd number greater than 8? A) 36 B) 49 C) 64 D) 81 E) 100

13. A gardener arranges tins of paint in a triangular stack: 1 tin in the top row, 2 in the next, 3 in the next, and so on. How many tins are in a stack with 6 rows?

14. What is $5^2 - 3^2$? A) 14 B) 15 C) 16 D) 17 E) 18

15. Explain, using an example, why a triangular number plus the next triangular number always gives a square number.

Special Numbers Challenge (15 questions)

1. Which is the first square number you reach if you start counting upward from 100?

2. If you list cube numbers in order, which is the biggest one still under 150?

3. Which of these square numbers also has a square root that is an even number? A) 25 B) 36 C) 49 D) 81

4. The 9th triangular number is 45. Without adding all the numbers again, work out the 10th triangular number.

5. Select ALL that apply: a number n is a triangular number, and $5 < n < 20$. Which of these values of n are correct? A) 6 B) 9 C) 10 D) 14 E) 15

6. The 5th triangular number is 15. Work out $5^2 + 5$, and check whether it equals double the 5th triangular number.

7. If a square-shaped playground covers 100 m^2 of ground, how long is each of its sides?

8. Which of these square numbers ends in the digit 6? A) 25 B) 36 C) 49 D) 81

9. Knowing that the 5th triangular number is 15, work out what the 6th and 7th triangular numbers must be.

10. Suppose $n^3 = 216$ for some whole number n. What must n be? A) 5 B) 6 C) 7 D) 8

11. Select ALL that apply: which of these statements about square and cube numbers are true? A) 1 is both a square number and a cube number. B) 64 is both a square number and a cube number. C) Every square number is positive. D) 8 is a square number. E) The square root of 49 is 7.

12. Tins are stacked in triangular layers: 1 tin on top, then 2, then 3, and so on. How many tins make up a stack 6 layers deep?

13. If n^2 works out to 144, what whole number is n? A) 11 B) 12 C) 13 D) 14

14. Only one whole number below 40 (other than 1) is both triangular and square at the same time. Which number is it?

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15. You know that adding up the first n odd numbers always gives n^2. Use this shortcut to find the total of the first 8 odd numbers.

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